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1 Wi-Fi, NR, and LTE Performance Comparison

1.1 Overview

This report summarizes FlowMonitor metrics collected from 72 deterministic ns-3 simulations covering Wi-Fi mesh (hotspot bands 5 GHz and 2.4 GHz), 5G NR, and LTE technologies. Each scenario varies STA/UE spawn mode (uniform vs anchored), STA/UE count (5, 10, 15), and application packet size (1 kB, 10 kB, 1 MB).

1.2 Common Configuration

- **Simulation domain:** 400 m × 400 m × 30 m
- **Mobility model:** Gauss-Markov for all STAs/UEs (speed 0.3–0.8 m/s, alpha 0.85, heights 0–30 m)
- **Infrastructure layout:**
 - Wi-Fi mesh: six-node topology with roaming STAs and hotspot APs
 - NR/LTE: two macro gNB/eNB sites at (−100, 200, 30) m and (500, 200, 30) m
- **Buildings:** static obstacles shared by all technologies using HybridBuildingsPropagationLossModel (frequency set per band)
- **Transmit settings:** hotspot/backhaul powers and antenna gains aligned across runs; Wi-Fi mesh hotspot/backhaul radios use up to 23 dBm, NR/LTE use 43 dBm (gNB/eNB) and 15 dBm (UE)
- **Traffic pattern:** HTTP/HTTPS/Video/TCP and VoIP flows initiated around 15 s into a 30 s simulation
- **Random seed:** Fixed seed to guarantee deterministic replay across all scenarios
- **Measurement pipeline:** FlowMonitor records per-flow statistics, which are parsed into the results/.../metrics.md summaries used for this report

1.3 Scenario Matrix

Dimension	Values
Technology	Wi-Fi 5 GHz, Wi-Fi 2.4 GHz, NR, LTE
Spawn mode	Uniform, Anchored
Client count	5, 10, 15 STAs/UEs
Packet size	1 kB, 10 kB, 1 MB

Total simulations: $3 \times 2 \times 3 \times 3 = 72$ (Wi-Fi counted twice for both hotspot bands).

1.4 Descriptive Statistics

Technology	Metric	Mean	Std Dev
Technology	Metric	Mean	Std Dev
Wi-Fi 5 GHz	PDR (%)	90.42	11.37
	Avg Delay (ms)	92.38	75.60
	Throughput (Mbps)	1.69	1.81
Wi-Fi 2.4 GHz	PDR (%)	90.97	11.19
	Avg Delay (ms)	103.10	90.23
	Throughput (Mbps)	1.87	2.06
NR	PDR (%)	100.00	0.00
	Avg Delay (ms)	15.92	0.00
	Throughput (Mbps)	0.02	0.00
LTE	PDR (%)	87.50	0.00
	Avg Delay (ms)	11.31	0.00
	Throughput (Mbps)	0.03	0.00

1.5 Scenario Comparisons

1.5.1 UE/STA Count = 5, Packet Size = 1 kB

Technology	Spawn Mode	PDR (%)	Avg Delay (ms)	Throughput (Mbps)
LTE	Anchored	87.50	11.31	0.03
LTE	Uniform	87.50	11.31	0.03
NR	Anchored	100.00	15.92	0.02
NR	Uniform	100.00	15.92	0.02
Wi-Fi 2.4 GHz	Anchored	99.05	68.23	4.29
Wi-Fi 2.4 GHz	Uniform	89.15	188.80	2.24
Wi-Fi 5 GHz	Anchored	99.98	113.82	3.35
Wi-Fi 5 GHz	Uniform	96.95	35.98	3.58

1.5.2 UE/STA Count = 5, Packet Size = 10 kB

Technology	Spawn Mode	PDR (%)	Avg Delay (ms)	Throughput (Mbps)
LTE	Anchored	87.50	11.31	0.03
LTE	Uniform	87.50	11.31	0.03
NR	Anchored	100.00	15.92	0.02
NR	Uniform	100.00	15.92	0.02
Wi-Fi 2.4 GHz	Anchored	99.61	70.13	7.76
Wi-Fi 2.4 GHz	Uniform	85.67	125.06	3.32
Wi-Fi 5 GHz	Anchored	99.42	114.55	5.14
Wi-Fi 5 GHz	Uniform	89.12	139.83	5.83

1.5.3 UE/STA Count = 5, Packet Size = 1 MB

Technology	Spawn Mode	PDR (%)	Avg Delay (ms)	Throughput (Mbps)
LTE	Anchored	87.50	11.31	0.03
LTE	Uniform	87.50	11.31	0.03
NR	Anchored	100.00	15.92	0.02
NR	Uniform	100.00	15.92	0.02
Wi-Fi 2.4 GHz	Anchored	100.00	4.39	0.03
Wi-Fi 2.4 GHz	Uniform	100.00	5.71	0.04
Wi-Fi 5 GHz	Anchored	100.00	4.60	0.03
Wi-Fi 5 GHz	Uniform	100.00	5.49	0.04

1.5.4 UE/STA Count = 10, Packet Size = 1 kB

Technology	Spawn Mode	PDR (%)	Avg Delay (ms)	Throughput (Mbps)
LTE	Anchored	87.50	11.31	0.03
LTE	Uniform	87.50	11.31	0.03
NR	Anchored	100.00	15.92	0.02
NR	Uniform	100.00	15.92	0.02
Wi-Fi 2.4 GHz	Anchored	94.42	72.21	1.50
Wi-Fi 2.4 GHz	Uniform	90.26	105.92	1.39
Wi-Fi 5 GHz	Anchored	93.74	103.16	1.49
Wi-Fi 5 GHz	Uniform	87.62	127.54	1.33

1.5.5 UE/STA Count = 10, Packet Size = 10 kB

Technology	Spawn Mode	PDR (%)	Avg Delay (ms)	Throughput (Mbps)
LTE	Anchored	87.50	11.31	0.03
LTE	Uniform	87.50	11.31	0.03
NR	Anchored	100.00	15.92	0.02

Technology	Spawn Mode	PDR (%)	Avg Delay (ms)	Throughput (Mbps)
NR	Uniform	100.00	15.92	0.02
Wi-Fi 2.4 GHz	Anchored	85.69	209.59	3.67
Wi-Fi 2.4 GHz	Uniform	85.50	153.70	2.01
Wi-Fi 5 GHz	Anchored	82.78	144.08	3.46
Wi-Fi 5 GHz	Uniform	85.72	102.98	2.54

1.5.6 UE/STA Count = 10, Packet Size = 1 MB

Technology	Spawn Mode	PDR (%)	Avg Delay (ms)	Throughput (Mbps)
LTE	Anchored	87.50	11.31	0.03
LTE	Uniform	87.50	11.31	0.03
NR	Anchored	100.00	15.92	0.02
NR	Uniform	100.00	15.92	0.02
Wi-Fi 2.4 GHz	Anchored	100.00	4.45	0.05
Wi-Fi 2.4 GHz	Uniform	100.00	4.55	0.04
Wi-Fi 5 GHz	Anchored	100.00	4.35	0.05
Wi-Fi 5 GHz	Uniform	100.00	4.53	0.05

1.5.7 UE/STA Count = 15, Packet Size = 1 kB

Technology	Spawn Mode	PDR (%)	Avg Delay (ms)	Throughput (Mbps)
LTE	Anchored	87.50	11.31	0.03
LTE	Uniform	87.50	11.31	0.03
NR	Anchored	100.00	15.92	0.02
NR	Uniform	100.00	15.92	0.02
Wi-Fi 2.4 GHz	Anchored	88.19	125.22	1.00
Wi-Fi 2.4 GHz	Uniform	65.89	266.16	0.70
Wi-Fi 5 GHz	Anchored	79.42	202.69	0.74
Wi-Fi 5 GHz	Uniform	71.81	155.83	0.49

1.5.8 UE/STA Count = 15, Packet Size = 10 kB

Technology	Spawn Mode	PDR (%)	Avg Delay (ms)	Throughput (Mbps)
LTE	Anchored	87.50	11.31	0.03
LTE	Uniform	87.50	11.31	0.03
NR	Anchored	100.00	15.92	0.02
NR	Uniform	100.00	15.92	0.02
Wi-Fi 2.4 GHz	Anchored	92.72	159.98	4.56
Wi-Fi 2.4 GHz	Uniform	61.27	284.25	0.98
Wi-Fi 5 GHz	Anchored	80.58	137.36	1.33
Wi-Fi 5 GHz	Uniform	60.50	258.73	0.85

1.5.9 UE/STA Count = 15, Packet Size = 1 MB

Technology	Spawn Mode	PDR (%)	Avg Delay (ms)	Throughput (Mbps)
LTE	Anchored	87.50	11.31	0.03
LTE	Uniform	87.50	11.31	0.03
NR	Anchored	100.00	15.92	0.02
NR	Uniform	100.00	15.92	0.02
Wi-Fi 2.4 GHz	Anchored	100.00	3.71	0.05
Wi-Fi 2.4 GHz	Uniform	100.00	3.81	0.06
Wi-Fi 5 GHz	Anchored	100.00	3.63	0.05
Wi-Fi 5 GHz	Uniform	100.00	3.72	0.06